Jacob W. Toney

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EDUCATION

Massachusetts Institute of Technology (MIT) **Cambridge, MA**

Ph.D., Chemical Engineering and Computational Science (Minor: Quantum Chemistry)
Doctoral Advisor: Prof. Heather J. Kulik
Thesis: Accelerating Quantum Chemistry with Machine Learning for the Rational Design of Transition Metal Complexes
Relevant Coursework: Computational Chemistry, Advanced Electronic Structure Theory, Machine Learning Algorithms, ML for Chemistry, Matrix Calculus for ML, Diffusion Models, Generative AI and Stochastic Differential Equations, Numerical Methods, Bayesian Modeling and Inference

University of Southern California (USC) **Los Angeles, CA**

B.S., Chemical Engineering (Minor: Materials Science) | GPA: 4.00 Aug. 2017 – May 2021
M.S., Chemical Engineering Sep. 2019 – May 2021
Research Advisor: Prof. Shaama M. Sharada
Thesis: "Investigating the Mechanisms of C–H Activation with Enzyme-Inspired Molecular Catalysts" (**Awarded Best Research Paper**)

PUBLICATIONS

1. G. G. Terrones, R. G. St. Michel, **J. W. Toney**, A. K. Ball, Y. Wang, A. G. Garrison, A. Nandy, R. Meyer, F. Edholm, C. Oh, S. G. Pujet, D. B. K. Chu, D. Muhammetgulyyev, H. J. Kulik, "molSimplify 2.0: Improved Structure Generation for Automating Discovery in Inorganic Molecular and Reticular Chemistry", *J. Chem. Inf. Model* **2026** (In Press).

2. S. Darouich, **J. W. Toney**, W. Luo, J. Kästner, M. Niepert, H. J. Kulik, "Beyond the Training Domain: Robust Generative Transition State Models for Unseen Chemistry", *arXiv*, **2026**, 10.48550/arXiv.2601.16469.

3. **J. W. Toney**, R. G. St. Michel, A. G. Garrison, I. Kevlishvili, H. J. Kulik, "Identifying Dynamic Metal–Ligand Coordination Modes with Ensemble Learning", *J. Am. Chem. Soc.* **2025**, 10.1021/jacs.5c17169.

4. **J. W. Toney**, R. G. St. Michel, A. G. Garrison, I. Kevlishvili, H. J. Kulik, "Graph neural networks for predicting metal–ligand coordination of transition metal complexes", *Proc. Natl. Acad. Sci.* **2025**, 122, 41, 10.1073/pnas.2415658122.

5. **J. W. Toney**, A. G. Garrison, W. Luo, R. G. St. Michel, S. Mukhopadhyay, and H. J. Kulik, "Exploring beyond experiment: generating high-quality datasets of transition metal complexes with quantum chemistry and machine learning", *Curr. Opin. Chem. Eng.* **2025**, 50, 10.1016/j.coche.2025.101189.

6. Y. Liu, **J. W. Toney**, J. M. Cavanagh, K. Sun, A. L. Smith, C. Yuan, R. G. St. Michel, P. A. Graggs, D. Toste, H. J. Kulik, T. Head-Gordon, "Exploring Transition Metal Complexes with Large Language Models", *ChemRxiv* **2025**, 10.26434/chemrxiv-2025-hm3zb.

7. I. Kevlishvili, R. G. St. Michel, A. G. Garrison, **J. W. Toney**, H. Adamji, H.-J. Jia, Y. Román-Leshkov, H. J. Kulik, "Leveraging natural language processing to curate the tmCAT, tmPHOTO, tmBIO, and tmSCO datasets of functional transition metal complexes", *Faraday Discuss.* **2024**, 256, 275-303, 10.1039/D4FD00087K.

8. Z. Lan*, **J. W. Toney***, S. M. Sharada, "A computational mechanistic study of CH hydroxylation with mononuclear copper-oxygen complexes", *Catal. Sci. Technol.* **2023**, 13, 342-351, 10.1039/D2CY01128J

HONORS & AWARDS

2025 Leslye Miller Fraser and Darryl M. Fraser Fellow, MIT School of Engineering, May 2025

2024 MathWorks Engineering Fellow, MIT Department of Chemical Engineering, June 2024

2022 Alfred P. Sloan UCEM Fellowship, MIT, September 2022

2021 Albert Dorman Valedictorian, USC Viterbi School of Engineering, May 2021
National Academy of Engineering Grand Challenges Scholar, USC Viterbi School of Engineering, May 2021
University Trustees Award (awarded for perfect GPA), USC, May 2021
Best Research Paper, "Bioinspired Catalysis of C–H Hydroxylation: A Computational Approach", USC Viterbi School of Engineering, May 2021
Best Analytical & Expository Essay, "Engineering Ethics: Grand Challenges and Grander Responsibilities", USC Viterbi School of Engineering, May 2021
Outstanding Chemical Engineering Student Award, USC Department of Chemical Engineering and Materials Science, May 2021
Best Research Presentation, "Investigating the Mechanisms of C–H Activation with Enzyme-Inspired Molecular Catalysts", USC Department of Chemical Engineering and Materials Science, February 2021

2019 Provost's Undergraduate Research Fellowship, USC, January 2019 – December 2020

2017 Presidential Scholar, USC, Aug. 2017
Latino Alumni Association Scholar, USC, Aug. 2017

PRESENTATIONS

1. American Institute of Chemical Engineers (AIChE), Boston, MA. "Graph neural networks for predicting metal–ligand coordination in transition metal complexes", talk in *Data Science and ML Approaches to Catalysis: Catalyst Design and Discovery*. November 2025.

2. World Association of Theoretical and Computational Chemists (WATOC), Oslo, Norway. "Graph neural networks for predicting metal–ligand coordination in transition metal complexes", talk in *Chemical Structure and Reactivity: Analysis*. June 2025.

3. CECAM Frontiers of Computational Reaction Prediction, Chicago, IL. "Graph neural networks for predicting metal–ligand coordination in transition metal complexes", poster. July 2024.

4. USC Department of Chemical Engineering and Materials Science Research Symposium, Los Angeles, CA (virtual). "Investigating the Mechanisms of C–H Activation with Enzyme-Inspired Molecular Catalysts", recorded talk. February 2021. **Awarded Best Research Presentation**.

5. AIChE Annual Student Conference, San Francisco, CA (virtual). "Computational Analysis of C–H Hydroxylation in Bio-inspired Monometallic Complexes", poster. November 2020.

6. USC Department of Chemical Engineering and Materials Science Research Symposium, Los Angeles, CA. "Computational Analysis of C–H Hydroxylation in Bio-inspired Monometallic Complexes", poster. March 2020.

7. AIChE Annual Student Conference, Orlando, FL. "Computational Analysis of C–H Hydroxylation in Bio-inspired Monometallic Complexes", poster. November 2019.

PROFESSIONAL EXPERIENCE

Aspen Technology **Bedford, MA**
Software Engineering Intern (MIT Practice School) Jul. 2023 – Aug. 2023

Woodside Energy **Perth, WA, Australia**
Engineering & Data Science (MIT Practice School) Jun. 2023 – Jul. 2023

SpaceX **Hawthorne, CA**
Operations Engineer (Full-time) Jun. 2021 – Aug. 2022

ExxonMobil Research & Technology Development **Houston, TX**
Process Engineering R&D Intern (Internship) Jun. 2020 – Aug. 2020

California Resources Corporation **Bakersfield, CA**
Facilities Engineering Intern: Gas Plants (Internship) May 2019 – Aug. 2019

TEACHING & COURSEWORK

5.698/10.637: Computational Chemistry (MIT) **Cambridge, MA**
Grader. Graduate elective in chemical engineering and chemistry. Sep. 2025 – present

10.34: Numerical Methods Applied to Chemical Engineering (MIT) **Cambridge, MA**
Graduate Teaching Fellow. Graduate chemical engineering core course. Sep. 2024 – Dec. 2024
Instructor rating: 6.5/7

10.65: Chemical Reactor Engineering (MIT) **Cambridge, MA**
Grader. Graduate chemical engineering core course. Feb. 2024 – May 2024

10.40: Chemical Engineering Thermodynamics (MIT) **Cambridge, MA**
Grader. Graduate chemical engineering core course. Sep. 2023 – Dec. 2023

MENTORSHIP & SERVICE

Undergraduate Students

1. Lauren Wright (Chemical Engineering, Louisiana State University)
   Project: Designing new hydroformylation catalysts through mechanistic study and data science.
   Presentation: AIChE Annual Student Conference, Boston, MA. "Leveraging Computational Chemistry and Data Science to Design Novel Hydroformylation Catalysts", poster. November 2025. **Awarded Third Place Poster**.

2. Sebastian Pujet (Chemical Engineering and Computer Science, Washington University in Saint Louis)
   Project: Active learning to optimize mechanochemical properties of transition metal complexes.
   Contributed Publications: G. G. Terrones, R. G. St. Michel, J. W. Toney, A. K. Ball, Y. Wang, A. G. Garrison, A. Nandy, R. Meyer, F. Edholm, C. Oh, S. G. Pujet, D. B. K. Chu, D.

Muhammetgulyyev, H. J. Kulik, "molSimplify 2.0: Improved Structure Generation for Automating Discovery in Inorganic Molecular and Reticular Chemistry", *J. Chem. Inf. Model* **2026** (In Press).

USC AIChE Student Chapter **Los Angeles, CA**
President Apr. 2020 – May 2021

USC Latino Alumni Association Scholar Committee **Los Angeles, CA**
President Jan. 2019 – May 2021